Flashing Process

1. Overview

The ESP32-S3 does not have MicroPython built in; on power-up it only runs the program already stored in Flash. Therefore, the first time you use it, you need to write the MicroPython firmware into the Flash — this process is called **flashing**.

Flashing essentially writes the `.bin` firmware file byte by byte into the chip's SPI Flash over the serial port. After flashing is complete, the ESP32-S3 automatically enters MicroPython REPL on every power-up, with no need to flash again.

This chapter introduces two flashing methods:

Method	Tool	Suitable Scenario
Command-line flashing	esptool (pip)	Development and debugging; can be scripted for batch operations
GUI flashing	Flash Download Tool	Beginners; visual configuration makes it hard to make mistakes

> For first-time use, try the GUI method first; after you are familiar with the parameters, use the esptool command line for better efficiency.

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2. Firmware Selection

2.1 Where to Download

MicroPython officially provides pre-compiled firmware for the ESP32-S3, so you do not need to compile it yourself:

[MicroPython ESP32-S3 download page]

2.2 Which Firmware to Choose

Firmware Type	Flash Requirement	PSRAM Requirement
`ESP32_GENERIC_S3`	≥ 8MB	None / ≤ 2MB Octal PSRAM
`ESP32_GENERIC_S3_SPIRAM`	≥ 8MB	≥ 8MB Quad/Octal PSRAM
`ESP32_GENERIC_S3_SPIRAM_OCT`	≥ 8MB	≥ 8MB Octal PSRAM

> The board used in this tutorial is the **ESP32-S3 N16R8 (16MB Flash + 8MB Octal PSRAM)**; select the `GENERIC_S3_SPIRAM_OCT` firmware.
>
> If you are not sure about the PSRAM type, download the `SPIRAM` firmware without the `_OCT` suffix (compatible with both Quad and Octal).

2.3 Version Selection

Download the latest **Stable** `.bin` file. The file name format looks like:

```
ESP32_GENERIC_S3-SPIRAM_OCT-20260406-v1.28.0.bin
|                   |          |          |
Chip platform   PSRAM/Flash config  Release date  Firmware version
```

> Avoid downloading the Nightly Build; it is not stable and is only suitable for trying out new features.

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3. Enter Flashing Mode

Before flashing, the chip must enter **Download Boot** mode so that the ROM Bootloader waits for serial data.

3.1 Automatic Entry

Most ESP32-S3 development boards support automatic downloading — `esptool` and Flash Download Tool automatically trigger download mode through the RTS/DTR pin timing, so you can just click to flash.

3.2 Manual Entry

If automatic entry fails (showing "Timed out waiting for packet header"), you need to do it manually:

```
Hold the BOOT key → press the EN key → release BOOT
```

Timing diagram:

```
BOOT  ──────┐      ┌──────────────────
            └──────┘  (hold)
EN    ──┐     ┌──────────────────────
        └─────┘  (press)
                 ← now entering download mode
```

> While holding BOOT, pressing EN resets to the Bootloader; the Bootloader detects BOOT being pulled low and enters download mode.

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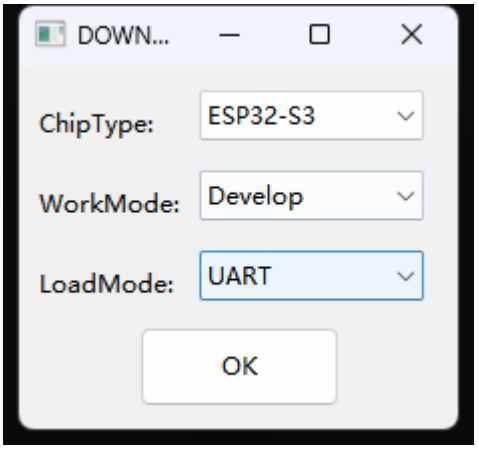
4. GUI Flashing (Flash Download Tool)

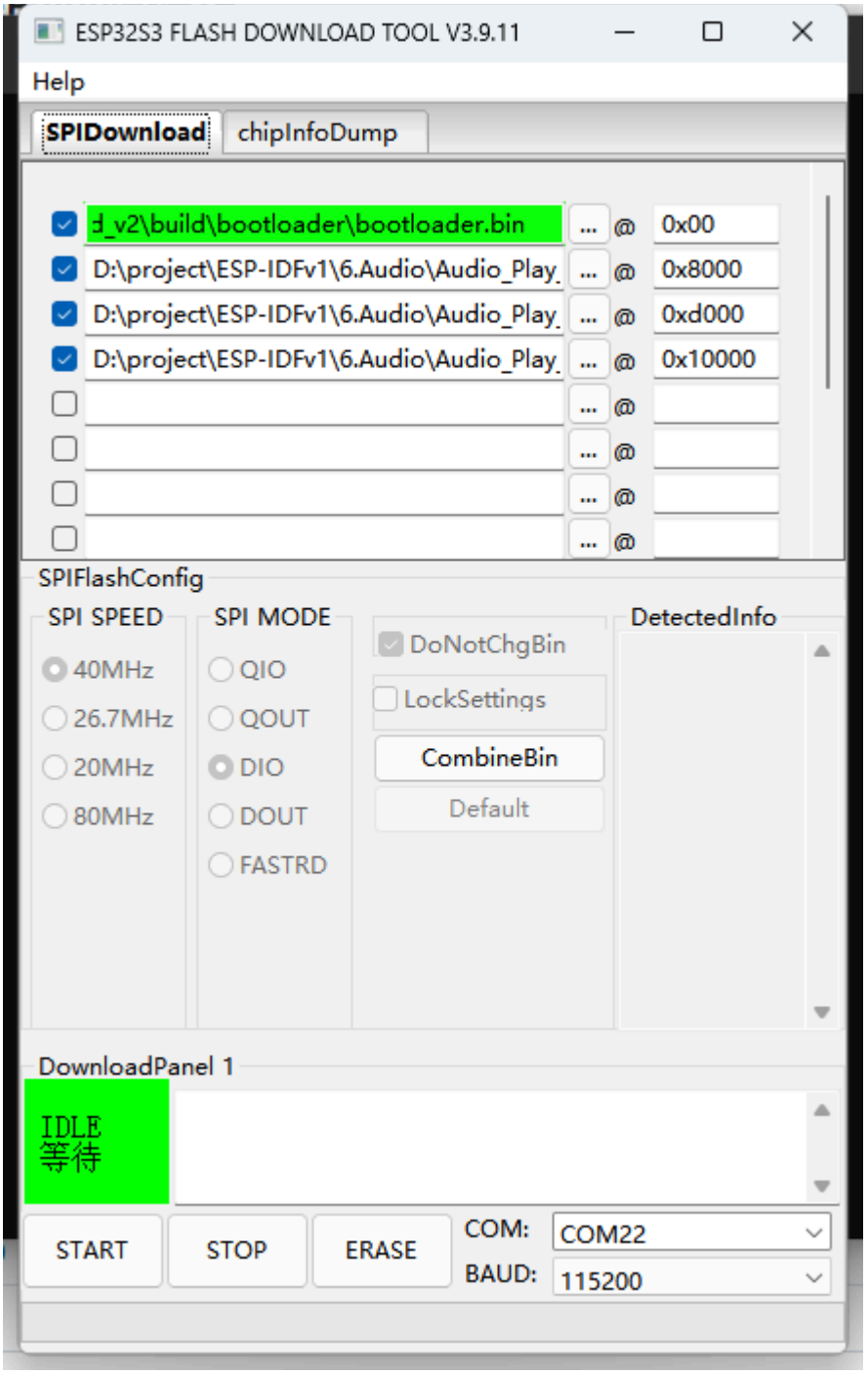
Flash Download Tool is Espressif's official Windows GUI tool, at `6.Software Tools → Firmware flashing tool → flash_download_tool → flash_download_tool_3.9.11.exe`.

4.1 Open the Tool

1. Double-click `flash_download_tool_3.9.11.exe`
2. Select **ESP32-S3** for chip type and **Develop** for work mode, then click **OK**



3. The main interface looks like this:



4.2 Configure Flashing Parameters

① Select the firmware file:

- Click the `...` browse button in the first row and select the downloaded MicroPython `.bin` file
- Make sure the start address in the first row is `0x0`

② SPI speed:

- Select `80MHz` (the default; good compatibility)

③ SPI mode:

- Select `DIO` (dual-line I/O; supported by most boards)
- If it fails to start after flashing, try `QIO` (quad-line I/O)

④ **Flash size:**

- Select `128Mbit` (= 16MB), corresponding to the ESP32-S3 N16R8

⑤ **COM port:**

- Select the corresponding port number (`COM3` / `COM4`, etc.)
- Use the default baud rate `115200` or increase it to `921600` for speed

> The higher the serial baud rate, the faster the flashing, but some USB-to-serial chips (especially CH340) are unstable at 921600. If it fails, drop back to 115200 and retry.

4.3 Start Flashing

1. Confirm the above configuration is correct
2. Confirm the board is connected and in download mode
3. Click the **START** button
4. When the progress bar completes, **FINISH** appears and the process is done

After flashing completes, press the **RST** key on the board to reset and enter MicroPython REPL.

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5. Command-Line Flashing (esptool)

`esptool` is Espressif's official Python CLI flashing tool. The Environment Setup chapter already covered the installation steps (`pip install esptool`). This section only summarizes the commands for quick reference.

5.1 Erase Flash

```
python -m esptool --chip esp32s3 --port COM3 erase-flash
```

5.2 Flash the Firmware

```
python -m esptool --chip esp32s3 --port COM8 write_flash -z 0x0
"D:\Micropython\ESP32_GENERIC_S3-SPIRAM_OCT-20260406-v1.28.0.bin"
```

Parameter	Description
`--chip esp32s3`	Target chip model
`--port COM8`	Serial port number (replace with your actual port)
`write-flash`	Write to Flash
`-z 0x0`	Firmware start address (ESP32-S3 starts at 0x0)

5.3 Flashing Multiple bin Files (Example)

Some scenarios require flashing multiple bin files at the same time, such as firmware + file system + partition table, using multiple address pairs:

```
esptool --chip esp32s3 --port COM3 write-flash -z \
    0x0        firmware.bin \
    0x8000     partition-table.bin \
    0x10000    littlefs.bin
```

Address	Content
`0x0`	Firmware (the MicroPython interpreter itself)
`0x8000`	Partition Table
`0x10000`	File system (LittleFS, stores .py scripts)

> The MicroPython single-bin firmware already includes the partition table and a blank file system, so normally you only need to flash the `0x0` address. Multiple bins are only used for custom partition scenarios.

5.4 View Flash Information

```
python -m esptool --chip esp32s3 --port COM7 flash_id
```

Example output:

```
Connected to ESP32-S3 on COM7:
Chip type:          ESP32-S3 (QFN56) (revision v0.2)
Features:           Wi-Fi, BT 5 (LE), Dual Core + LP Core, 240MHz, Embedded PSRAM 8MB
(AP_3v3)
Crystal frequency:  40MHz
MAC:                a4:cb:8f:c6:b8:64

Stub flasher running.

Flash Memory Information:
=========================
Manufacturer: 46
Device: 4018
Detected flash size: 16MB
Flash type set in eFuse: quad (4 data lines)
Flash voltage set by eFuse: 3.3V

Hard resetting via RTS pin...
```

```
C:\Users\Administrator>python -m esptool --chip esp32s3 --port COM7 flash_id
Warning: Deprecated: Command 'flash_id' is deprecated. Use 'flash-id' instead.
esptool v5.3.1
Connected to ESP32-S3 on COM7:
Chip type:          ESP32-S3 (QFN56) (revision v0.2)
Features:           Wi-Fi, BT 5 (LE), Dual Core + LP Core, 240MHz, Embedded PSRAM 8MB (AP_3v3)
Crystal frequency:  40MHz
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Hard resetting via RTS pin...
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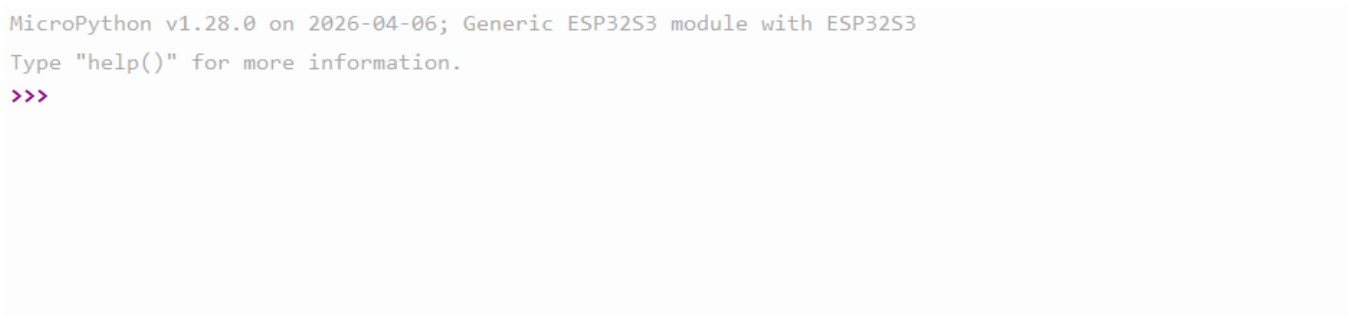
6. Verify the Flashing Result

After flashing and resetting, use a serial tool to verify:

6.1 Verify with a Serial Assistant

Open any serial assistant (baud rate 115200), connect to the corresponding COM port, press RST to reset, and you should see:

```
MicroPython v1.28.0 on 2026-04-06; ESP32S3 module with ESP32S3
Type "help()" for more information.
>>>
```



6.2 Verify with Thonny REPL

Connect the ESP32 with Thonny; if `>>>` appears in the Shell, it is normal.

6.3 Troubleshooting Verification Failures

Symptom	Possible Cause
No output on the serial port	Wrong baud rate; firmware not flashed successfully
Garbled output	Baud rate mismatch (MicroPython defaults to 115200)

Symptom	Possible Cause
Output shows `waiting for download`	Accidentally entered download mode and did not reset; press EN to exit
Repeated restarts outputting `rst:0x1`	Flash or PSRAM configuration mismatch (wrong firmware selected)

7. Common Problems

Symptom	Cause	Solution
Flash Download Tool cannot find the COM port	Driver not installed	Install the CH340 driver and reopen the tool
Flashing progress bar stuck	The chip did not enter download mode	Manually enter download mode with BOOT + EN
Cannot start after flashing	Firmware does not match the chip	Confirm the firmware is an `ESP32_GENERIC_S3` series
`esptool` reports "command not found"	The Python Scripts directory is not in PATH	Close and reopen the terminal, or use `python -m esptool`
Flashing succeeds but PSRAM is unavailable	Firmware does not include PSRAM support	Switch to a `SPIRAM` version firmware
Flash Download Tool shows "COM port occupied"	The serial port is occupied by another program	Close the serial assistant, Thonny, Arduino Monitor, etc.
Files in the file system lost after multiple flashes	Flashing erased the entire Flash	Back up the `.py` files on the device via Thonny before flashing